



Society of Petroleum Engineers

SPE-229677-MS

A Method for Characterizing Fluid Contacts in Freshwater Environments Exposed to Extrinsic Heat Utilizing Time-Spectral Pulsed Neutron Logging Measurements

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Copyright 2025, Society of Petroleum Engineers DOI [10.2118/229677-MS](https://doi.org/10.2118/229677-MS)

This paper was prepared for presentation at the ADIPEC held in Abu Dhabi, UAE, 3 – 6 November 2025.

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Abstract

Identifying freshwater pay zones using electrical resistivity logging presents challenges primarily due to the minimal resistivity contrast between hydrocarbons and freshwater. Prior research has documented the application of thermal energy to a formation to modify the characteristics of freshwater, reporting variations in resistivity measurements that facilitate the identification of fluid contacts through resistivity logging. Further exploration of an alternative measurement with distinct physics leads us to propose the utilization of pulsed neutron logs derived from time spectra when the formation temperature is elevated through an additional heating source.

When a formation is subjected to non-intrinsic heat, such as targeted heating through electromagnetic irradiation, the properties of fluid components occupying the pores exhibit variations. The in-situ density change in water exceeds that observed in oil (and sometimes gas, depending on the reservoir pressure). Multidetector pulsed neutron well logging acquires time-spectral data and provides gamma-ray ratios from various combinations of multiple scintillation detectors. Two gamma-ray ratio measurements sensitive to fluid density variations (or hydrogen index) have been developed: (1) inelastic scattering of fast neutron-induced gamma-ray ratio and (2) thermal neutron capture-induced gamma-ray ratio measurements.

The MCNP models reveal that these ratio measurements are particularly effective in indicating the conversion of freshwater into steam. The inelastic ratio log delineates steam formation as temperature increases, accompanied by a reduction in its response. Similarly, the capture ratio measurement also exhibits a decrease in its response; however, this reduction is comparatively less pronounced than that observed in the inelastic ratio measurement.

The conventional macroscopic thermal neutron capture cross section, sigma log, demonstrates its sensitivity in the transition from freshwater to steam; however, it does not provide the same level of resolution for the freshwater-to-steam transition as the ratio measurements. Carbon/Oxygen (C/O) logging can also indicate the interfaces between hydrocarbons and freshwater; however, the sensitivity of the C/O ratio is significantly dependent on porosity and hole size. Consequently, challenges persist in identifying fluid contacts using the C/O ratio in low-porosity formations with large holes.

This innovative method for precisely delineating pay zones in freshwater environments will assist petrophysicists and reservoir engineers, particularly during the exploration phase of reservoir fluid characterization. Recent enhancements of slimhole multidetector pulsed neutron well logging instruments have facilitated the faster logging of these ratio measurements. Additionally, the heat-flasked pulsed neutron tool permits pulsed neutron data acquisition in harsh temperatures exceeding 350 degrees Fahrenheit.

Introduction

The identification of formation fluid contacts is critical from the well construction to the hydrocarbon production phase of the field. Methods for delineating formation fluid types vary depending on the well condition and formation fluid specifications. During the well construction phase or before the well completion phase (i.e., when the hole is barefoot), resistivity measurement is generally used to differentiate between water and hydrocarbon zones, particularly when water salinity is high. However, if the formation water salinity is low and variable, the resistivity log is not effective. The technology for identifying hydrocarbon and water contacts is limited when the formation water salinity becomes low (Eltaher et al., 2018).

Nuclear magnetic resonance (NMR) logging or formation testing and sampling (FT&S) can provide formation fluid contacts in a freshwater environment. However, these methods are time-consuming compared to resistivity logging. Alternatively, dielectric permittivity measurement is a viable approach in low-salinity water environments. The dielectric technology can determine hydrocarbon saturation in any formation water salinity (Forgang et al., 2019). This is particularly useful in situations where conventional resistivity-based methods are unreliable due to nonconductive formation water or unknown water resistivity. However, the depth of investigation of dielectric measurement (approximately 8 inches) is shallower than that of resistivity logs. Moreover, all existing NMR, FT&S, and dielectric technologies are typically operated with a rig on location in newly drilled wells or as part of a workover operation, mainly because the sizes of the tools are too large to pass through production tubing for routine rigless reservoir monitoring.

Considering freshwater environment challenges, previous studies focused on heating the formation to change the resistivity responses of the target formations (Al-Huwaider & Ma, 2023a, 2023b). The lab test results revealed that the resistivity property of water changed when the freshwater was exposed to heat. It was reported that the changes in resistivity with varying temperature depend on water saturation, with a reduction in resistivity showing a direct relationship to water saturation. This study suggested a method to alter the resistivity of fluids and identify fluid contacts. (Al-Hamad et al., 2024).

Building upon the concept of exposing extrinsic heat to the formation, an exploration of an alternative measurement method was undertaken, specifically, pulsed neutron measurements. This paper discusses pulsed neutron well logging, measurement physics, and stochastic simulation results to elucidate the applicability of pulsed neutron logs for identifying formation fluid contacts, including gas, oil, and freshwater.

Pulsed Neutron Well Logging Technology

Background of Pulsed Neutron Logging Tool

Pulsed neutron well logging is most commonly used to evaluate properties of formations behind casing or tubing strings. Surveillance of reservoirs for all categories of open-hole-completed and cased wells has been conducted through the application of diverse pulsed neutron well logging techniques and data interpretation methods. This includes quantification of two- or three-phase clastic or carbonate reservoir fluids in all ranges of water salinity environments. Furthermore, a nuclear production logging application is available using pulsed neutron logging, facilitating the mitigation of certain limitations associated with conventional

centerline or array production logging sensors in challenging downhole logging environments. (Chace et al., 1994; Trcka et al., 1996; and Kim et al., 2022).

The outer diameter of the first commercially introduced pulsed neutron tool for the oil and gas industry, equipped with a single scintillation detector, was 3-5/8 inches (Youmans et al., 1964). With the advancement of pulsed neutron logging instruments, the first slimhole multidetector pulsed neutron tool whose outer diameter is 1-11/16 inches was introduced in 1999 (Gilchrist et al., 1999). The outer diameter of the pulsed neutron tool is smaller than that of resistivity, neutron porosity, and bulk density logging tools, resulting in a less restrictive well logging tool. This feature is especially beneficial for through-tubing formation evaluation.

Key components of a pulsed neutron tool comprise a pulsed neutron generator, a scintillation detector and its crystal material, and gamma-ray processing electronics. The pulsed neutron generator emits 14.2 megaelectron volt (MeV) high-energy neutrons, and the neutron outputs are controllable. Unlike radioactive chemical neutron porosity tools, which are generally used in open holes, the higher energy level and controllability of neutrons allow pulsed neutron logging to be widely used in cased wells. Fast neutrons generated by the pulsed neutron generator penetrate borehole fluids, tubing, casing, and cement sheaths to the formations and interact with formation elements. A gamma-ray is released as a result of the interaction of a neutron and the nuclei of the element. Induced gamma-rays are generated via inelastic scattering of fast neutrons with specific nuclei, thermal neutron capture events, or neutron activation. Induced gamma-rays are recorded by multiple scintillation detectors (i.e., short-, long-, and extra-long-spaced detectors from the closer proximity of the pulsed neutron generator), and gamma-ray counts are registered via the pulsed neutron tool's electronics. The gamma-rays are measured based on their arrival time relative to the pulsed neutron source burst (i.e., generating time-spectrum data for each detector) and based on the energy level of the induced gamma-ray (i.e., creating energy-spectrum data for each detector).

The multidetector pulsed neutron tool is capable of recording time spectra or energy spectra, depending on the acquisition mode. This multifunctional data acquisition system was enhanced to enable the latest version of the pulsed neutron tool to acquire not only time spectra or energy spectra, but also simultaneous data acquisition of time and energy spectra. The latest version of the pulsed neutron tool, the next-generation pulsed neutron tool, was upgraded from the previous-generation pulsed neutron tool for the three main key components. The enhancement includes (1) a pulsed neutron generator that produces twice as many fast neutrons as the legacy tool, (2) the adoption of a denser scintillation detector crystal (lanthanum bromide, LaBr_3), which enables improved gamma-ray stopping power, thereby resulting in higher gamma-ray counting efficiency and providing higher peak resolution, and (3) digital electronics that reduce dead time, thus increasing the gamma-ray processing efficiency (Nardiello et al., 2022; Kim et al., 2023a; Akagbosu et al., 2023; and McGlynn et al., 2023).

Additionally, there is an industry initiative and effort to make pulsed neutron measurements intrinsic rather than tool-dependent, so data from different tools and service providers can be compared and used consistently (Ma et al., 2021a, 2021b). Extracting formation water salinity (Ma et al., 2014; Mostefai et al., 2024) and characterizing formation mineralogy (Van Steene et al., 2023) using pulsed neutron technologies are also key areas of interest.

High-Temperature Pulsed Neutron Logging Tool

The application of pulsed neutron logging for this study is conducted in a high-temperature environment. The operating temperature limit for a standard pulsed neutron tool is 350 degrees Fahrenheit; therefore, a heat-flasked pulsed neutron tool developed for logging in harsh temperature conditions is necessary for this type of application.

The specifications of the heat-flasked pulsed neutron tool include a thermal isolator, a distributed heat sink, and a special electronics chassis (Kim et al., 2021, 2023a). The outer diameter of the high-temperature tool is 2.55 inches. The heat-flask housing facilitates the even dissipation of heat, preventing heat shock to

the electronics. This allows for logging for approximately 12 hours, with the borehole temperature reaching as high as approximately 500 degrees Fahrenheit.

Time-Spectra-Based Pulsed Neutron Data

The acquisition of gammarays based on time provides a time spectrum for each of the three detectors. From the moment the pulsed neutron generator emits fast neutrons, induced gamma rays are recorded with respect to the time of arrival at the detectors. While fast neutrons are generated during the burst cycle, the inelastically scattered neutrons mainly induce gamma rays, and some thermalized neutrons also induce gamma rays. Conversely, when the source is off and the fast neutrons have been thermalized, neutron capture events produce gamma rays. Therefore, depending on the time domain in which gamma rays are recorded in the time spectra, it is possible to distinguish between inelastic or capture-based pulsed neutron measurements.

The characteristics of inelastic and capture pulsed neutron measurements mainly differ in their depth of investigation and their dependence on water salinity and hydrogen index. The representative time spectra-based inelastic pulsed neutron measurement, widely used for formation evaluation, is a ratio of inelastic gamma-ray count rates from the short- and extra-long-spaced detectors over the period of 0–100 microseconds (hereafter denoted as RIN13). Note that the pulsed neutron generator's burst period is 60 microseconds within each source pulsing cycle. The equivalent ratio measurement, which involves gamma-ray count rates from capture events over the period of 200–400 microseconds for the short-spaced-detector-based time spectrum and 200–1,000 microseconds for the extra-long-spaced-based time spectrum, is RATO13.

The depths of investigation for RIN13 and RATO13 are 6–10 inches and 12–16 inches, respectively. RIN13 is essentially insensitive to salinity compared to RATO13. With the same changes in water salinity, RIN13 remains only minimally affected by variations in chlorine concentration and its effect on hydrogen density, whereas RATO13 is highly dependent on water salinity. Due to this salinity dependency, RATO13 is used in saline water environments where water salinity is known and high (Conner et al., 2018), whereas RIN13 can be used in environments where salinity is low, high, or unknown (Itter et al., 2015). Both RIN13 and RATO13 are sensitive to hydrogen index variation, but RIN13 exhibits a much greater sensitivity to hydrogen index changes. The hydrogen index dependency of RIN13 also enables the quantification of formation gas and the identification of gas or gas-condensate reservoir depletion. In addition, RIN13 is more sensitive to lithology changes (matrix point variations between sandstone, limestone, and dolomite, for example) while RATO13 shows minimal sensitivity between limestone and dolomite. The distinctions between RIN13 and RATO13 have been extensively documented (Inanc et al., 2009; Kim et al., 2013; 2016; 2017; 2021; 2022; 2023a; 2023d; 2024a; 2024b; 2025).

A macroscopic thermal neutron capture cross section (formation sigma) measurement, which is the first pulsed neutron measurement employed for saline water saturation analysis, is also obtained with RIN13 and RATO13. Formation sigma is recognized as one of the most statistically reliable measurements, and it offers the greatest depth of investigation among all pulsed neutron well logs utilized for formation evaluation. Nevertheless, formation sigma primarily detects changes in salinity, and its sensitivity to differentiate freshwater, oil, methane, and steam is limited (Kim et al., 2021). Consequently, formation sigma is not incorporated into this study.

An important aspect of pulsed neutron measurements based on time spectra is their enhanced statistical stability when compared to data derived from energy spectra. The gamma-ray count rates associated with time spectra are inherently higher than those observed in energy spectra (including inelastic, capture, or background energy spectra), which causes logs obtained from time spectra to exhibit a superior signal-to-noise ratio. This results in slower statistical errors in time-based pulsed neutron logs compared to those based on energy spectra. In addition, the data acquisition time for time-based measurements is faster than that for energy-based measurements. For instance, pulsed neutron capture (PNC) logging using

time spectra from three detectors, producing RIN13 and RATO13, can be logged at 20 feet per minute (ft/min) with the next-generation pulsed neutron tool, whereas carbon/oxygen (C/O) logging (Kim et al., 2023a,2023b) is performed at 6 ft/min and typically requires multiple passes to reduce measurement uncertainty. The uncertainty in formation evaluation utilizing C/O is also highly influenced by porosity. The lower the porosity, the greater the uncertainty in the evaluation results (Kim et al., 2023c). Consequently, this study focuses on the measurements obtained from time-spectra-based inelastic and capture logs (i.e., RIN13 and RATO13), as the efficiency of characterizing a freshwater formation contact improves with shorter data acquisition durations.

Thermal Energy Transfer to Formations

Various methods have been studied to transfer heat to formations and alter the properties of formation fluids. Thermal energy application technology is primarily employed in the oil and gas industry for enhanced oil recovery (EOR) processes, which effectively increase the mobility of heavy oil and thereby improve the hydrocarbon recovery factor. The thermal EOR techniques include steam injection (such as cyclic stimulation and steam flooding), hot water flooding, well stimulation, in-situ combustion, and electrical heating. Electrical heating methods include low-frequency, low-medium frequency, and high-frequency heating. Low-frequency heating uses electrical resistance heating, low-medium frequency heating employs induction heating, and high-frequency heating involves microwave and radio frequency methods.

As an example of a low-frequency heating application, Temizel et al. discussed electrical resistance heating in multilateral wells for the recovery of heavy oil (Temizel et al., 2020). Conversely, regarding high-frequency heating applications, Bientinesi et al. introduced an innovative radiofrequency/microwave (RF/MW) technique that integrates a downhole antenna with an interface structure (tight shell) between the radiating well and the reservoir (Bientinesi et al., 2013). This design aimed to attain more uniform heating of the reservoir while restricting elevated temperatures in the vicinity of the wellbore. The study demonstrated that water vaporization exerts a substantial influence on temperature profiles and facilitates (a) the limitation of temperature increase near the wellbore region, and (b) a more effective distribution of irradiated energy within the reservoir. The experimental results indicate that, after approximately 28 hours of irradiation at 1,000 watts, temperatures in certain sections of the oil sand mass surpassed 100°C, signifying the commencement of water vaporization.

In considering an optimal configuration for the pulsed neutron tool tailored to this application, a pulsed neutron logging tool equipped with a heating source and a temperature sensor is deemed ideal. The heating source within the pulsed neutron tool provides thermal energy to the formation, thereby elevating the formation temperature and facilitating the conversion of freshwater into steam. Once the formation water has been heated and the formation becomes saturated with steam, the pulsed neutron logging components immediately acquire time-spectral data. When the pulsed neutron tool is equipped with a heating source complemented by a temperature sensor, there is no requirement for an additional trip to deploy the pulsed neutron tool after utilizing a device to heat the formations. The acquired pulsed neutron logs (i.e., RIN13 and RATO13) are capable of distinguishing between oil and freshwater contacts. This application can be used during the well construction phase, notably during drilling or well placement stages.

Challenges in heating formation and converting water to steam remain regarding power requirements, heat dissipation, and wellbore integrity (Pandian et al., 2020). Electrical heating necessitates a dependable and adequate power supply, which may present challenges in remote locations. Heat dissipation from the heated zone, particularly in heterogeneous reservoirs, can necessitate more sophisticated heating strategies. Maintaining the integrity of wellbores and heating elements during extended operation is essential. Consequently, the implementation of this study is primarily dependent on the availability and practicality of heat generation within the formations. Its application may be limited to shallow formations.

Nuclear Modeling-Based Evaluation of Pulsed Neutron Logging Application

We used the Monte Carlo N-Particle (MCNP) transport code, customized for oil and gas well logging, to generate the predicted pulsed neutron tool responses.

Background of MCNP Modeling

The utilization of MCNP for modeling in pulsed neutron well logging has been demonstrated to be an effective method for simulating the interactions of neutrons with various materials within a geological formation. MCNP modeling facilitates a comprehensive simulation of neutron transport, inelastic scattering, and capture interactions with the surrounding elements, including formation fluid constituents such as oil, gas, and water.

The modeling process begins with the creation of a geometric model of the wellbore and surrounding formation. This involves defining parameters such as borehole and completion specifications, borehole fluid properties, and formation lithology and fluid characteristics to establish an accurate simulation environment. Once the geometry is constructed, a neutron source pulsing is simulated. MCNP modeling enables the investigation of these interactions in a stochastic manner, providing statistical distributions of gamma-ray outputs based on defined well configurations and formation properties. The results establish a theoretical foundation for pulsed neutron logs, specifically tailored to the modeling parameters, thereby offering a valuable tool for assessing the feasibility of pulsed neutron logging applications.

MCNP Modeling Parameters

We assumed the heat exposure for the shallow formations, where the formation pressure is 1,000 psi and the water salinity is zero parts per million (ppm) NaCl equivalent. The well specifications, formation lithology, and type of pulsed neutron tool used for modeling are shown in [Table 1](#).

Table 1—MCNP modeling parameters

Openhole size	6.125 inches
Pulsed neutron tool type	Next-generation heat-flasked pulsed neutron tool
Casing size	None
Fluid in the hole	Water-based mud (WBM) with a density of 11.5 pounds per gallon (ppg)
Lithology	Limestone

We estimated the densities of water, oil, and methane as a function of reservoir temperature, as summarized in [Table 2](#) and displayed in [Fig. 1](#). These estimates of water densities were based on the correlation suggested by Wagner and Pruß ([Wagner & Pruß, 2002](#)). The boiling point of water was calculated to be 284.81 °C. Oil densities were determined using the Standing correlation (1947) for an oil characterized by a 25-degree API, a gas-oil ratio of 100 scf/STB, and a gas specific gravity of 0.55. The reservoir pressure is assumed to be greater than the bubble point pressure.

At a temperature of 0 °C and a pressure of 1,000 psi, the densities of water, oil, and methane are calculated as 1.003 g/cc, 0.907 g/cc, and 0.1018 g/cc, respectively. As the temperature rises, all three fluids exhibit a decrease in density. The extent of this decline is illustrated in [Fig. 1 \(b\)](#). During the initial 100 °C increase, the reduction in oil density surpasses that of water or methane. When the temperature increase reaches 200 °C, the densities of oil and water decrease similarly, whereas the decline in gas density is comparatively limited. Water transitions into steam at a temperature of 284.81 °C under a pressure of 1,000 psi. It is essential to note that at this temperature, water can exist in either a liquid or a vapor state. Further temperature increases result in substantial density reductions, primarily due to the phase change from liquid to vapor. The density of methane exhibits comparatively less variation than that of water or oil. Furthermore, when water

transitions to steam, its density approximates that of methane, indicating that densities in the gaseous phase are similarly low, within the range of 0.0246–0.0359 g/cc at this particular pressure (refer to Fig. 1(a)).

Table 2—Densities of water, oil, and methane with variations of temperature at a pressure of 1,000 psi

Temperature (°C)	Density (g/cc)			
	Water	Steam	Oil	Methane
0	1.0030	N/A	0.9070	0.1018
100	0.9615		0.8239	0.0563
200	0.8687		0.7532	0.0416
284.81	0.7416	0.0359	0.6975	0.0350
300	N/A	0.0332	0.6880	0.0341
400		0.0246	0.6290	0.0305

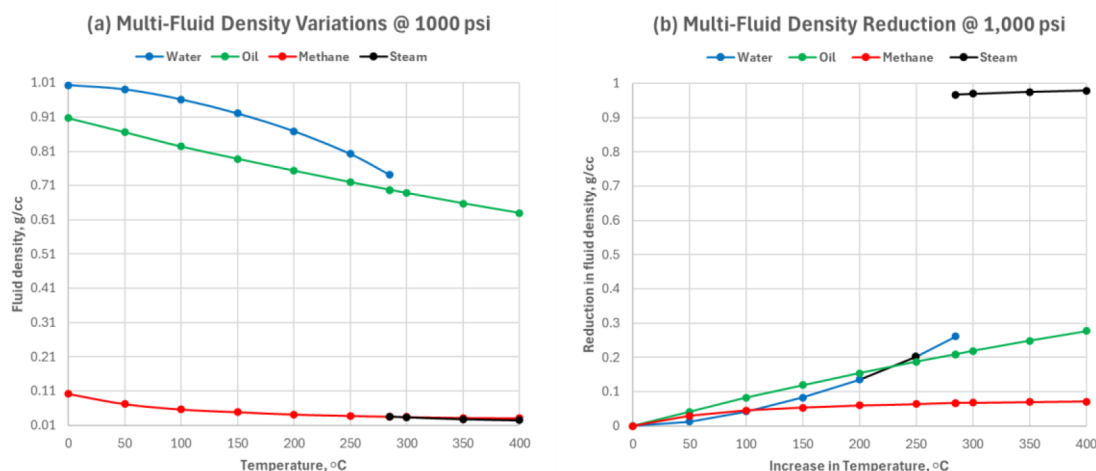


Figure 1—Reservoir fluid densities and their variations: (a) estimated in situ formation water, oil, and gas densities with respect to variations in reservoir temperature at a pressure of 1,000 psi; and (b) the reduction in fluid densities with increasing temperature.

Stochastic Nuclear Modeling Results

MCNP modeling results of RIN13 and RATO13 are presented in Figs. 2 and 3. For all figures, the horizontal axis represents limestone porosity in fractions, and the vertical axis shows RIN13 responses with (a) water- or steam-filled limestones, (b) oil-filled limestones, or (c) methane-filled limestones. Multiple lines in each plot represent pulsed neutron measurements for limestones occupied by fluids with six varying densities.

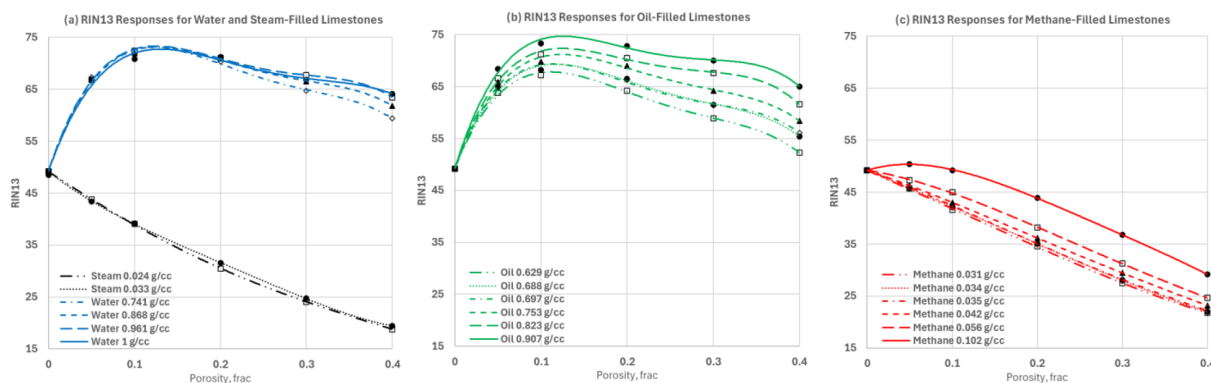


Figure 2—MCNP modeling results for RIN13 of (a) water- and steam-filled formations, (b) oil-filled formations, and (c) methane-filled formations at six temperatures at a pressure of 1,000 psi.

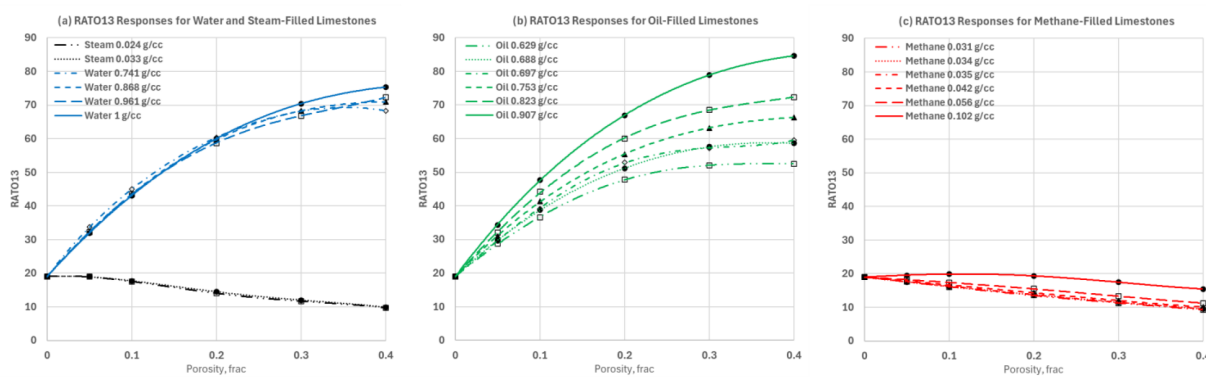


Figure 3—MCNP modeling results for RATO13 of (a) water- and steam-filled formations, (b) oil-filled formations, and (c) gas-filled formations at six temperatures at a pressure of 1,000 psi.

Inelastic Gamma-Ray Ratio (RIN13) Responses. When the completion specifications and formation lithology remain unchanged, the responses of RIN13 for the liquid-filled formation surpass those for the methane- or steam-filled formation, owing to a higher hydrogen index. As water density varies from 1 g/cc to 0.741 g/cc, corresponding to temperature changes from 0 °C to 284.81 °C at 1,000 psi, the phase of water remains liquid. Consequently, the hydrogen index is elevated compared to gaseous-phase fluids, indicating that the responses of RIN13 are higher than those of any methane-filled limestone cases (Figs. 2(a) and 2(c)). Furthermore, the responses of RIN13 remain comparable with the reduction in water density caused by an increase in temperature within the liquid phase. Similarly, the variation in oil density, although resulting in a greater magnitude change in RIN13 compared to water density variation, does not significantly affect RIN13 declines to the level of methane-filled limestone (Fig. 2(b)).

As presented in Fig. 2(c), when the pores of limestone are filled with methane, RIN13 exhibits a comparatively lower value than that observed in liquid-filled formations. The hydrogen index of the gas is lower than that of the liquid, resulting in distinctly suppressed RIN13 responses. A decrease in gas density from 0.102 g/cc to 0.056 g/cc (representing a reduction of 0.046 g/cc, or a 45% decrease in formation gas density) leads to a noticeable decrease in RIN13. The magnitude of RIN13 reduction from 0.102 g/cc to 0.056 g/cc methane-filled limestones is most significant compared to the reductions associated with further declines in gas density (or increases in temperature), such as from 0.056 g/cc to 0.031 g/cc (a reduction of 0.025 g/cc). The responses of RIN13 for methane-filled limestone indicate a direct relationship with gas density (or the hydrogen index of the gas); however, this relationship is not linear.

As the temperature exceeds the boiling point of 284.81 °C at 1,000 psi, the RIN13 for steam-filled limestones declines to levels below those of the methane-filled limestone responses. This RIN13 response

facilitates the identification of freshwater-filled formations when they are subjected to heating, resulting in the conversion of freshwater into steam.

Capture Gamma-Ray Ratio (RATO13) Responses. Fig. 3 presents MCNP modeling results for RATO13.

The major trends of responses of RATO13 are the same as those of RIN13. As the water salinity is zero, RATO13 does not reveal salinity dependency, meaning that four water lines with varying in-situ densities are fairly similar. When the temperature surpasses the boiling point and creates steam in the pores, RATO13 is suppressed (refer to responses in black lines of Fig. 3(a)). As the densities of oil and methane decrease, the respective RATO13 responses decrease (refer to Figs. 3(b) and 3(c)). This behavior is identical to that observed in RIN13. However, the magnitude of changes in RIN13 and RATO13 with respect to variations in fluid density differs.

Fig. 4 presents the maximum variations of RIN13 and RATO13 for the highest and lowest densities of each fluid type (water or steam, oil, and methane). These variations (or dynamic ranges) indicate the formation sensitivity of the measurements for the two different density fluid-filled formations.

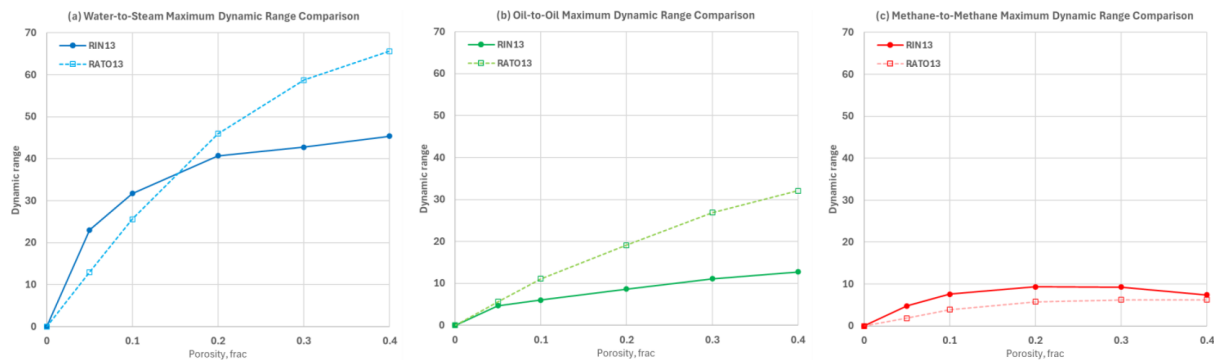


Figure 4—The maximum dynamic ranges of RIN13 and RATO13 for the respective fluids of the largest and smallest densities: (a) a water-to-steam-filled limestones case; (b) a case for limestones with densities of 0.907 g/cc and 0.629 g/cc for oils; and (c) a case for limestones filled with 0.102 g/cc and 0.031 g/cc of methane.

Referring to Fig. 4(a), which shows formation sensitivity between 1 g/cc water and 0.024 g/cc steam-filled limestones for RIN13 and RATO13, the dynamic range between water- and steam-filled limestones for RATO13 is higher than that for RIN13 when porosity exceeds 0.15 (fraction). This indicates that RIN13 is more effective than RATO13 for detecting the transition from water to steam at porosity levels below 0.15. Conversely, when porosity exceeds 0.15, RATO13 is more effective in indicating the water-to-steam phase change. In general, RATO13 has a greater depth of investigation than RIN13, thereby presenting a wider range of formation sensitivities. However, RIN13 has a unique characteristic in that its water-to-steam dynamic range expands to a much greater magnitude at low porosity than that of RATO13. Both RIN13 and RATO13 exhibit characteristics of sufficient decline in values when a water phase change occurs; thus, the two measurements can be used to identify the phase change, which ultimately means delineating the freshwater-filled formations after sufficient exposure to extrinsic heat.

When oil density changes from 0.907 g/cc to 0.629 g/cc, the variations of RIN13 and RATO13 are shown in Fig. 4(b). The dynamic range for RATO13 exceeds that for RIN13 because RATO13 is more sensitive to changes in oil properties (i.e., due to thermal neutron capture by hydrogen). In any porosity range up to 0.4, the dynamic range for RATO13 remains greater than that for RIN13 when pores are filled with two different oil densities. The difference in dynamic range caused by oil density changes is less significant than that caused by water-to-steam variations (comparison of Figs. 4(a) and 4(b)).

Differences of RIN13 and RATO13 for methane-filled formations with two gas densities (i.e., 0.102 g/cc and 0.031 g/cc) are compared in Fig. 4(c). The difference is greater with RIN13 than with RATO13 because

the sensitivity in hydrogen index variation is higher for RIN13 than for RATO13 (Nardiello et al., 2017; Kim et al., 2023d). RIN13 exhibits increased sensitivity to variations in gas density.

Comparison of Responses for Steam- and Methane-Filled Formations. Fig. 5 shows the variation in RIN13 and RATO13 relative to changes in the saturated gaseous-phase types and densities, specifically the in-situ densities of methane and steam. Six methane densities, as detailed in Table 2, are used to simulate the responses of RIN13 and RATO13 for limestone filled with steam.

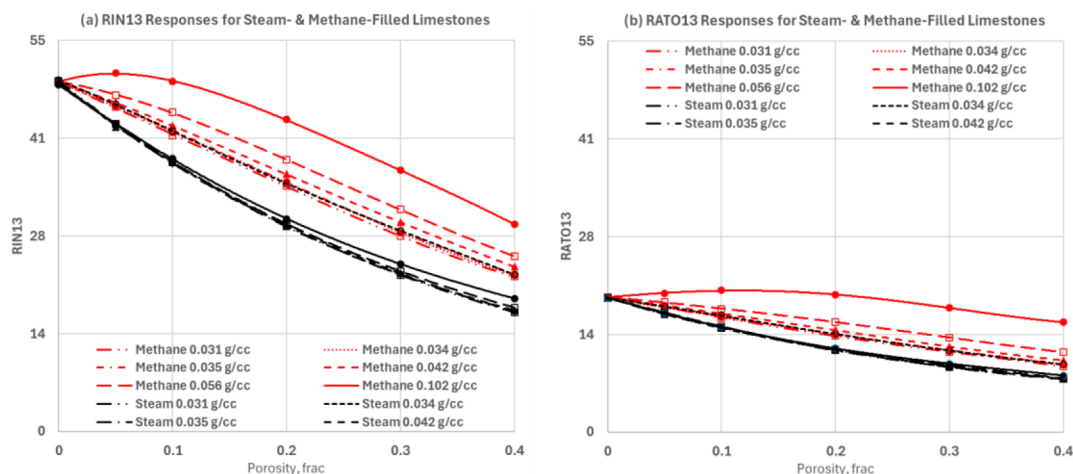


Figure 5—Comparison of RIN13 and RATO13 for limestones saturated with steam and methane.

At the identical gaseous-phase density, the responses of RIN13 and RATO13 are reduced for steam compared to methane-saturated formations. This discrepancy is attributable to the lower hydrogen density present in steam relative to methane at equivalent densities.

Delineation of Gas, Oil, and Freshwater Fluid Contacts

The objective is to identify the formation fluid contacts, such as gas-liquid contact (GLC) or oil-water contact (OWC). To evaluate this goal, we generated RIN13 and RATO13 log responses assuming 10, 20, and 30 pu limestones, and the same thickness of gas, oil, and freshwater-filled limestones.

Figs. 6, 7, and 8 present the RIN13 and RATO13 logs with temperature changes at three porosities—10, 20, and 30 pu, respectively. Gas-, oil-, and water- or steam-filled limestones correspond to the intervals of 0–100 ft, 100–200 ft, and 200–300 ft, respectively. Logs in red, orange, yellow, green, blue, and purple represent RIN13 or RATO13 responses when formation temperatures are 0°C, 100°C, 200°C, 284.81°C, 300°C, and 400°C, respectively. Note that in all cases, the distinction between methane- and oil-formations or between methane- and freshwater-filled formations does not require formation heating.

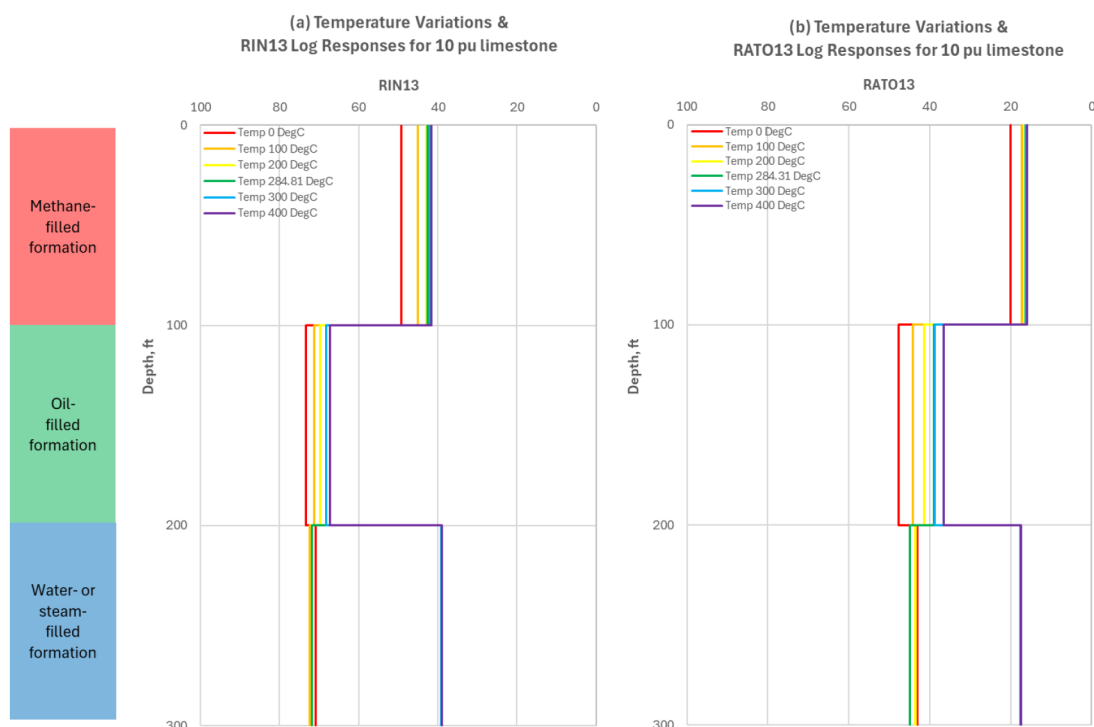


Figure 6—Log Responses of (a) RIN13 and (b) RATO13 concerning temperature variation for methane-, oil, and water- or steam-saturated 10 pu limestones.

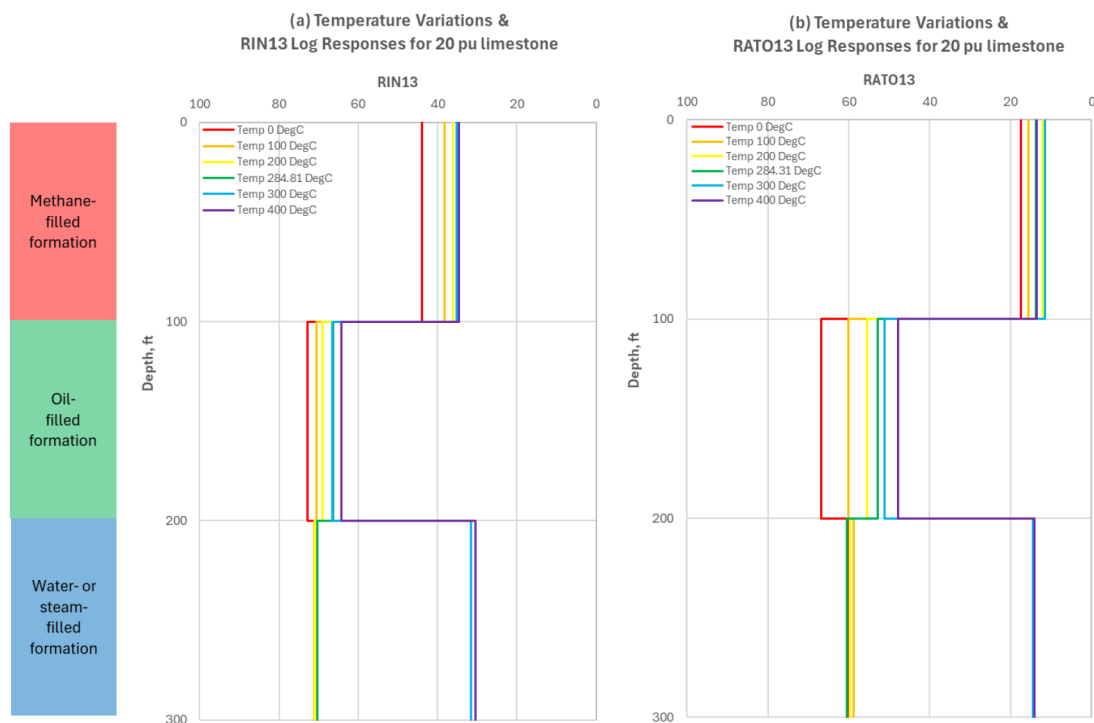


Figure 7—Log Responses of RIN13 and RATO13 concerning temperature variation for methane-, oil, and water- or steam-saturated 20 pu limestones.

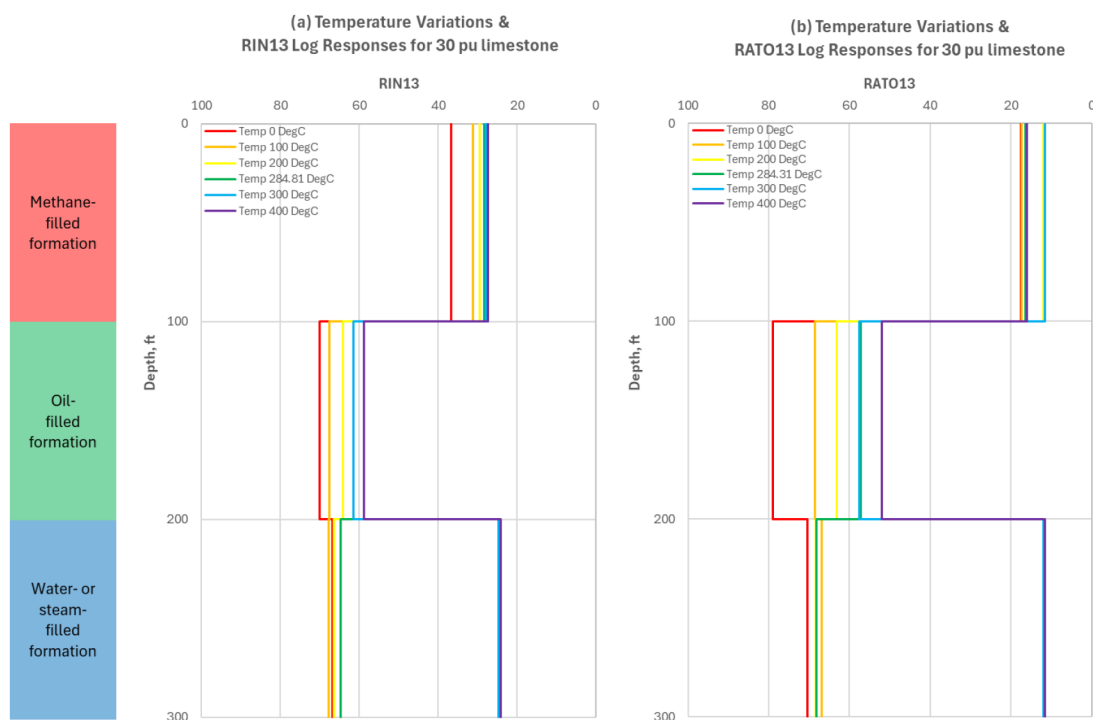


Figure 8—Log Responses of RIN13 and RATO13 concerning temperature variation for methane-, oil, and water- or steam-saturated 30 pu limestones.

Log Responses for a 10 pu-Limestone Case

As presented in Fig. 6(a), at any temperature, RIN13 responses for the methane-filled formation are lower than those for the liquid-filled formations. When the freshwater is converted to steam above the boiling point temperature, the responses of RIN13 for the 200–300 ft interval are lower than those for the methane-filled and oil-filled formations (i.e., the 0–100 ft and 100–200 intervals). This log presentation features that the formation is filled with water at the boiling point temperature and is filled with steam at temperatures of 300 °C and 400 °C. This log response allows the determination of an oil-freshwater contact at 200 ft in the post-heat treatment stage.

Similarly, in Fig. 6(b), the responses of RATO13 concerning temperature variations indicate the OWC when freshwater is converted to steam at a depth of 200 feet. As the temperature increases, the variations in RATO13 for the oil-filled formation (the interval of 100–200 ft) are more pronounced than those for the methane-filled formation (the 0–100 ft interval). Conversely, the fluctuations in RIN13 with temperature changes are more significant for the methane-saturated formation than for the oil-filled formation. As discussed earlier, this characteristic is attributed to the fact that RIN13 is more responsive to the hydrogen index than RATO13, and that RATO13 exhibits higher sensitivity to changes in oil properties.

Log Responses for a 20 pu-Limestone Case

The comparison of RIN13 and RATO13 logs from the 10 pu limestone case and the 20 pu limestone case indicates that RIN13 and RATO13 decrease as porosity increases in methane- or steam-filled formations. Additionally, the variation in measurements for the oil-filled limestone increases, particularly more pronounced in RATO13 than in RIN13. The principle of identifying the freshwater contact at the 20 pu limestone case is feasible.

Log Responses for a 30 pu-Limestone Case

As porosity increases from 20 pu to 30 pu, RIN13 and RATO13 exhibit lower values for methane- and steam-filled formations, while the separation between RIN13 and RATO13 for the oil-filled formation broadens.

It is observed that the increased porosity leads to a wider separation in RIN13 and RATO13 between water- and steam-filled limestone, as well as between oil- and steam-filled limestone. Therefore, limestone with a larger porosity is a favorable condition to implement this application.

Summary and Conclusions

We evaluated the use of pulsed neutron well logging to determine the contact between hydrocarbons and freshwater in the formation. The application of pulsed neutron logging is feasible when formation water is converted into steam after vaporization via external heat exposure.

The primary discussions and conclusions are summarized as follows:

- As formations are exposed to thermal treatment, the temperature at downhole conditions is generally greater than that of the standard pulsed neutron well logging tool operating limit. A pulsed neutron tool that can withstand harsh temperature conditions is required. For this study, we evaluated the use of a heat-flasked next-generation pulsed neutron tool.
 - The heat-flasked pulsed neutron tool is equipped with a special heat-circulating housing comprising a thermal isolator, a distributed heat sink, and a special electronic chassis.
 - Additionally, the next-generation pulsed neutron tool enables faster logging of pulsed neutron data acquisition compared to legacy pulsed neutron tools. This was enabled with the recent advancement of pulsed neutron well logging instrument upgrades.
 - The latest upgrades include a pulsed neutron generator that produces twice as many fast, high-energy neutrons, denser scintillation detector crystal material for higher count rates and higher spectral resolution, and digital electronics that can efficiently handle the higher gamma ray rates.
 - The combination of heat-flasked housing and upgrades on key pulsed neutron tool components enables the use of pulsed neutron technology to achieve the objective of identifying hydrocarbon and freshwater contact after the heat treatment.
- Time-spectra-based pulsed neutron measurements can be acquired faster than energy-spectra-based data. Additionally, time spectra are statistically more stable than energy spectra. Thus, logs extracted from time spectra have been studied.
 - RIN13 is an inelastic pulsed neutron log derived from short- and extra-long-spaced spectra, whereas RATO13 is a capture pulsed neutron log derived from the identical spectra yet later time domains.
 - RIN13 and RATO13 can be recorded at 20 ft/min with a heat-flasked next-generation pulsed neutron well logging tool.
- We performed a series of MCNP modeling for a 6.125-inch openhole completed, water-based-mud-filled well across limestone formations. The temperature variation ranged from 0 °C to 400 °C at a pressure of 1,000 psi. The boiling point was estimated to be 284.81 °C.
- The formations saturated with methane, oil, and water were identified with RIN13 and RATO13.
 - The responses of RIN13 and RATO13 were suppressed in limestones saturated with steam. Therefore, an oil-water contact can be identified once the formation temperature reaches or exceeds the boiling point of the water.
 - The identification of a gas-oil contact or a gas-water contact does not require formation heating.
 - The formation sigma log is also obtained during the recording of RIN13 and RATO13. The resolution of formation sigma distinguishing freshwater from steam is inadequate, as compared to RIN13 and RATO13. Furthermore, the formation sigma log does not provide a sufficient

contrast between oil and methane. The responses for freshwater, steam, oil, and methane are similar, rendering it unsuitable for this application.

- C/O logging is a salinity-insensitive method capable of distinguishing between freshwater and oil; however, the measurements are derived from inelastic energy spectra. Consequently, the logging speed is typically slower, approximately 3 to 10 times slower, depending on the type of pulsed neutron tool, compared to time-spectral data acquisition.

We demonstrated that pulsed neutron logging can be used to identify the hydrocarbon-freshwater contact. Further research on formation heating technology is necessary to enable the application of pulsed neutron well logging technology for delineating formation fluid contacts in freshwater environments. A modification of the pulsed neutron tool, such as the addition of a heating source and a temperature sensor, could be considered a prospective topic for future investigation.

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